Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science andCommerce

Department of Information Technology (B.Sc.I.T Semester V)

AI

Practical – III

Roll No.: T001	Name: Soham Acharekar
Class: TYIT	Batch: 01
Date of Assignment: 18/7/26	Date/Time of Submission:18/7/26

A medical test is used to detect a particular disease.The probability that a randomly selected person has the disease is 1%.If a person has the disease, the test returns positive with probability 99%.If a person does not have the disease, the test still returns positive withprobability 5% (false positive rate).A person takes the test and receives a positive result.

CODE:

```
# Probability of having disease
P_D = 0.01
# Probability of positive test if disease exists
P_Pos_D = 0.99
# Probability of positive test if disease does not exist
P_Pos_NotD = 0.05
# Probability of not having disease
P_NotD = 1 - P_D
# Bayes Rule
P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))
print("Probability that person actually has disease:")
print(round(P_D_Pos * 100, 2), "%")
print("Soham Acharekar T001")
```

OUTPUT:

```
Probability that person actually has disease:
16.67 %
Soham Acharekar T001
```